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Applied Physiology of Marathon Running

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Summary

Performance in marathon running is influenced by a variety of factors, most of which are of a physiological nature. Accordingly, the marathon runner must rely to a large extent on a high aerobic capacity. But great variations in maximal oxygen uptake ($\dot{V}O_{2\,max}$) have been observed among runners with a similar performance capacity, indicating complementary factors are of importance for performance. The oxygen cost of running or the running economy (expressed, e.g. as $\dot{V}O_{2\,15}$ at 15 km/h) as well as the fractional utilisation of $\dot{V}O_{2\,max}$ at marathon race pace ($\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$) [where Ma = mean marathon velocity] are additional factors which are known to affect the performance capacity. Together $\dot{V}O_{2\,max}$, $\dot{V}O_{2\,15}$ and $\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$ can almost entirely explain the variation in marathon performance. To a similar degree, these variables have also been found to explain the variations in the 'anaerobic threshold'. This factor, which is closely related to the metabolic response to increasing exercise intensities, is the single variable that has the highest predictive power for marathon performance. But a major limiting factor to marathon performance is probably the choice of fuels for the exercising muscles, which factor is related to the $\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$. Present indications are that marathon runners, compared with normal individuals, have a higher turnover rate in fat metabolism at given high exercise intensities expressed both in absolute (m/sec) and relative ($\%\dot{V}O_{2\,max}$) terms. The selection of fat for oxidation by the muscles is important since the stores of the most efficient fuel, the carbohydrates, are limited. The large amount of endurance training done by marathon runners is probably responsible for similar metabolic adaptations, which contribute to a delayed onset of fatigue and raise the $\dot{V}O_{2Ma} \times \dot{V}O_{2max}^{-1}$. There is probably an upper limit in training kilometrage above which there are no improvements in the fractional utilisation of $\dot{V}O_{2\,max}$ at the marathon race pace. The influence of training on $\dot{V}O_{2\,max}$ and, to some extent, on the running economy appears, however, to be limited by genetic factors.

The increased interest during the last two decades in long-distance running and especially marathon running has created unique opportunities for thorough studies on the physiology of the endurance capacity of runners. This has resulted in an increasing amount of data from different scientific fields concerning important variables for performance capacity in the marathon runner. To make a brief but complete review of this large field is hardly possible. We have therefore found it preferable to examine some of these interesting and most important areas in greater detail. Thus, we have concentrated on estimates of how marathon performance is related to such physiological variables as:

1) Maximal oxygen uptake ($\dot{V}O_{2\,max}$).

2) Oxygen cost of running at submaximal velocities, generally referred to as running economy (e.g. $\dot{V}O_{2\,15}$ at 15 km/h).

3) Fractional utilisation of $\dot{V}O_{2\,max}$ during running (e.g. $\dot{V}O_{2\,15} \times \dot{V}O_{2\,max}^{-1}$).

4) Various ways of measuring the metabolic response at increasing exercise intensities, such as the exponential increase of blood lactate concentration. This kind of metabolic response has been used to define a parameter commonly referred to as the 'anaerobic threshold'. The predictive power of this variable will be discussed.

5) Because of the duration and the intensity of the race we have also found it necessary to consider the quantity and quality of available fuels and the regulation of their utilisation during the race.

6) In much less detail, we shall discuss the influence of some environmental and other pertinent factors such as dehydration.

7) Finally, we have estimated the effects of endurance training.

To further elucidate the inter-relationship between marathon performance and different functional variables, we have included unpublished data from our laboratory on 35 marathon runners (Sjödin and Svedenhag, unpublished data).

Because of a lack of adequate data on female marathon runners, we shall not consider possible sex differences. Recovery from a marathon race and nutritional aspects of marathon running are examples of other areas that will not be discussed. If the reader desires additional information, we recommend other review articles, such as those by Costill and Miller (1980), Mahler and Loke (1984), Pate and Kriska, (1984) and Locksley (1980).

1. Central and Peripheral Physiological Characteristics of Marathon Runners

1.1 Efficient Cardiovascular Systems

Marathon running imposes great demands on the cardiovascular system and the employed locomotor organs. Centrally, an efficient oxygen transport system thus seems to be vital for success in marathon running, and enlarged heart volumes and well-developed vascular systems have been noted in successful endurance athletes, including marathon runners (Keul et al., 1982). Calculations based on echocardiographic measurements have also indicated larger diastolic volumes and a somewhat thicker wall of the left ventricle in marathon runners compared with controls (Parker et al., 1978; Paulsen et al., 1981). These enlarged dimensions of the heart have been suggested to be important for the cardiac pump performance, in order to achieve a large stroke volume and cardiac output which is thought to be the main determinant of the maximal oxygen uptake (Blomqvist and Saltin, 1983). A high capacity of the central circulation is, of course, important during such prolonged exertion as marathon running, and elite marathon runners have been estimated to use as much as 92% of their maximal cardiac output for 2.1 to 2.5h and to utilise up to 95% of their maximal stroke volume during competition (Costill, 1972). Cardiovascular regulation is not static, however, during the course of the effort since it has been noted that the stroke volume declines gradually with time while the heart rate increases (Costill, 1972; Saltin, 1964).

1.2 Muscle Fibre Types

There is a mixture of 2 main fibre types in the human skeletal musculature, type I [slow twitch (ST)] muscle fibre and type II [fast twitch (FT)] muscle fibre (Padykula and Herman, 1955). Type I fibres are characterised by a relatively higher oxidative capacity as indicated by high respiratory capacity (Ivy et al., 1980) and high activities of oxidative enzymes (Essén et al., 1975). Type II fibres have a relatively higher glycolytic potential and are more fatiguable than type I fibres (Tesch, 1980; Thorstensson, 1976). With endurance training the oxidative capacity of both types of fibre increases. The difference in activities between the 2 fibre types with regard to the pattern of Krebs cycle enzyme activities has been reported to be similar (Essén-Gustavsson and Henriksson, 1984) or smaller (Chi et al., 1983; Jansson and Kaijser, 1977) in the trained than in the untrained state. However, with regard to the enzymes of the fat oxidative pathway the difference between fibre types may be similar whether or not the individual is endurance-trained (Chi et al., 1983; Essén-Gustavsson and Henriksson, 1984). Differences between the 2 main fibre types have also been reported regarding the distribution of the different isozymes of lactate dehydrogenase (LDH).

Accordingly, the relative activity of heart-specific LDH (H-LDH), which preferentially oxidises lactate to pyruvate, has been found to increase specifically in slow twitch fibres with endurance training (Sjödin, unpublished results). This will increase the difference between the fibre types in the musculature of endurance-trained individuals.

Type I muscle fibres are predominant in the active musculature of marathon runners. A comparison between elite long-distance (mainly marathon) runners, elite middle-distance runners and untrained men showed a significantly higher portion of type I muscle fibres in the gastrocnemius muscle of the long-distance runners (79%) compared with corresponding values for the other two groups (MD 62% and UM 58%, $p < 0.05$). In addition, the oxidative potential, as measured by the activity of the mitochondrial enzyme succinate dehydrogenase, was significantly higher ($p < 0.05$) in the long-distance group (Costill et al., 1976). In our unpublished material we also found a significantly higher proportion of type I fibres in elite and good runners compared with slow runners (table I). The relative distribution of type I muscle fibres in the active leg musculature (vastus lateralis) has also been found to have some positive relationship with performance capacity in marathon running (Sjödin and Jacobs, 1981). A corresponding positive relationship was also found for the capillary density in muscle tissue. Thus, a predominance of type I fibres in the active muscles may be an advantage for performance in long-distance running, including the marathon. This advantage in endurance may be related to a higher capacity to oxidise fat and lactate in this type of fibre, with an economising effect on carbohydrate stores (see below).

Having given this general background, we shall now turn to the specific variables determining marathon performance.

2. *Physiological Factors Related to Marathon Performance*

2.1 Maximal Oxygen Uptake

For several decades it has been known that the maximal oxygen uptake ($\dot{V}O_{2\,max}$) is high in endurance event athletes (Åstrand, 1955; Robinson et al., 1937; Saltin and Åstrand, 1967). Consequently, marathon runners are expected to have high $\dot{V}O_{2\,max}$ values. This has been confirmed in several studies. Thus, in elite marathon runners (here defined as athletes with personal records below 2h 30 min), the maximal oxygen uptake has been found to be 70.9 ml/kg/min (n = 10, mean personal best 2:23; Costill and Fox, 1969; Costill and Winrow, 1970), 74.2 ml/kg/min (n = 5, 2:16; Svedenhag and Sjödin, 1984), 74.1 ml/kg/min (n = 8, 2:15; Pollock, 1977) and 79.0 ml/kg/min (n = 13, 2:13; Davies and Thompson, 1979).

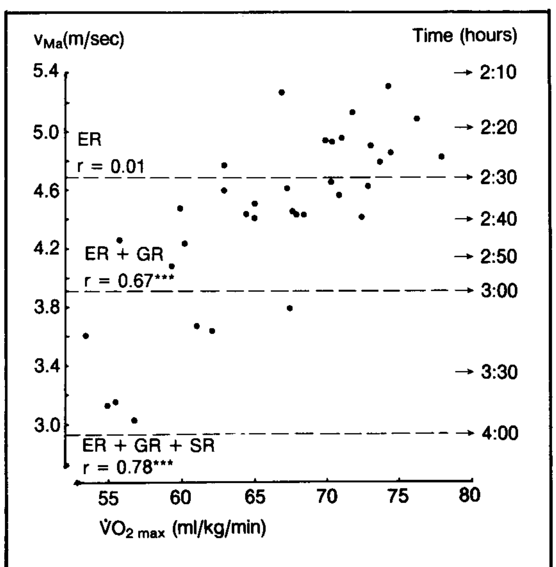

Fig. 1. The relationship between the individual mean marathon velocity (v_{Ma}) and $\dot{V}O_{2\,max}$ in elite runners (ER) [n = 12], in ER + good runners (GR) [n = 28] and in ER + GR + slow runners (SR) [n = 35]. *** = $p < 0.001$.

The importance of a high $\dot{V}O_{2\,max}$ for a high performance capacity in the marathon is also shown by the significantly different values of $\dot{V}O_{2\,max}$ for marathon runners at different levels of performance (table I) and by the correlation of r = 0.78 between $\dot{V}O_{2\,max}$ and marathon race pace (fig. 1). Similar correlations (r = 0.63 to 0.91) have been reported previously (table II).

Table I. Physical and physiological characteristics of the different categories of marathon runners (mean values ± SE)

	Elite runners (n = 12) < 2:30 (mean 2:21)	Good runners (n= 16) 2:30 - 3:00 (mean 2:37)	Slow runners (n = 7) > 3:00 (mean 3:24)
Age (years)	25.5 ± 1.3 (20-35)	29.9 ± 1.9 (18-44)	35.9 ± 1.8[a,d] (27–42)
Weight (kg)	65.8 ± 2.0 (47.9-72.7)	67.0 ± 2.0 (52.8-80.2)	70.9 ± 3.1 (62.7-85.0)
Type I fibres[e] (%)	76.0 ± 4.8 (52-99)	63.7 ± 5.2 (35-78)	56.0 ± 4.4[a] (36-66)
Years of training	7.0 ± 1.3 (1-15)	4.1 ± 0.5[b] (1-9)	2.4 ± 0.9[a] (1-7)
Number of marathon races	4.9 ± 1.5 (0-18)	4.3 ± 1.2 (0-19)	0.7 ± 0.4 (0-3)
Average weekly kilometrage	145 ± 9 (110-214)	115 ± 6[b] (76-156)	57 ± 10[a,c] (27-95)
$\dot{V}O_{2\,max}$ (ml/kg/min)	71.8 ± 1.2 (62.9-77.9)	65.6 ± 1.2[a] (55.8-72.8)	58.7 ± 1.9[a,c] (53.4-67.4)
$\dot{V}O_{2\,15}$ (ml/kg/min)	45.4 ± 0.7 (41.3-49.4)	48.6 ± 0.6[b] (43.6-51.6)	51.4 ± 1.6[a,d] (47.5-60.0)
$\dot{V}O_{2\,15} \times \dot{V}O_{2\,max}^{-1}$(%)	63.5 ± 0.8 (59.8-70.4)	74.3 ± 1.1[a] (67.7-83.8)	87.7 ± 1.7[a,c] (79.7-93.9)
$\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$ (%)	80.0 ± 1.0 (73.4-83.5)	80.4 ± 0.7 (73.9-83.6)	71.0 ± 2.7[a,c] (58.7-79.0)
$V_{Ma} \times V_{Hla\,4}^{-1}$ (%)	92.8 ± 0.8 (87.3-97.0)	92.1 ± 0.6 (86.7-96.1)	84.8 ± 2.1[a,c] (78.5-90.1)
$\dot{V}O_{2\,Hla\,4} \times \dot{V}O_{2\,max}^{-1}$(%)	87.9 ± 0.8 (83.6-92.9)	88.3 ± 0.6 (84.5-93.2)	84.8 ± 1.5[b,c] (79.2-89.8)
$V_{Hla\,4}$ (m/sec)	5.37 ± 0.05 (5.05-5.75)	4.85 ± 0.05[a] (4.40-5.13)	4.04 ± 0.05[a,c] (3.86-4.18)

a Significantly ($p < 0.01$) different from elite runners.
b Significantly ($p < 0.05$) different from elite runners.
c Significantly ($p < 0.01$) different from good runners.
d Significantly ($p < 0.05$) different from good runners.
e Biopsies were taken from 8, 7 and 6 runners in the elite, good and slow group, respectively.

NB. For statistical evaluation, a one-way analysis of variance was employed (ANOVA). The least significance group difference corresponding to a significance level of $p < 0.05$ or $p < 0.01$ was calculated for those variables which had a significant F-ratio (Snedecor and Cochran, 1968).

Although important, the maximal oxygen uptake is only one of the factors that determine success in the marathon. This is illustrated by the large variation in performance between runners of equal $\dot{V}O_{2\,max}$ and vice versa (fig. 1). Consequently, when a subgroup of runners with more similar performance capacities are studied, there may be no correlation between marathon performance and $\dot{V}O_{2\,max}$ (Costill, 1972) (fig. 1, e.g. for runners below 2:30, r = 0.01, n.s.). Furthermore, in some top-class marathon athletes, relatively low $\dot{V}O_{2\,max}$ values have been noted (Costill et al., 1976). Costill et al. (1971) reported a value of 69.7 ml/kg/min in a former 'world-record' holder (Clayton, 2:08.33), and we have noted a $\dot{V}O_{2\,max}$ of 66.8 ml/kg/min in the runner (Ståhl) who finished fourth in the 1983 world championship competition. In addition, when the $\dot{V}O_{2\,max}$ of elite marathon runners is compared with that of elite athletes in middle- and long-distance running, lower values may be found for the marathon runners (Pollock, 1977; Svedenhag and Sjödin, 1984). Thus, it would seem that although a high maximal oxygen uptake is certainly of great importance for marathon performance, it may not be as important as in runners specialising in long-distance running events of shorter duration, e.g. 5000 and 10,000 metres. Evidently, other factors can compensate for a relatively low $\dot{V}O_{2\,max}$ (see below).

Table II. Correlations between marathon performance and the maximal oxygen uptake

Authors	n	Performance range	Correlation
Foster et al. (1977)	23	2:23 - 4:08	r = 0.86
Farrell et al. (1979)	13	2:17 - 3:49	r = 0.91
Hagan et al. (1981)	50	2:19 - 4:58	r = 0.63
Maughan and Leiper (1983)	18	2:19 - 4:53	r = 0.88
Sjödin and Svedenhag (unpublished)	35	2:12 - 3:52	r = 0.78

All correlations were significant.

The cause of such a suggested difference in $\dot{V}O_{2\,max}$ between marathon runners and runners of shorter events is not known. It could be due to differences in training, e.g. more long-distance training at moderate intensities by marathon runners. The fact that marathon runners are often older than the runners of shorter distances may also partly explain such a difference since $\dot{V}O_{2\,max}$ declines with age (Åstrand and Rodahl, 1977). Alternatively, it may be due to particular characteristics of runners who choose the marathon. The duration of the marathon race makes it necessary to consider additional factors such as the capacity to oxidise fat and delay glycogen depletion (see below), which may contribute to a high performance capacity in the marathon but does not appear to be important in shorter long-distance events.

2.2 Running Economy

It is a well-established fact that the submaximal oxygen requirements of running at a specific speed may vary considerably between subjects (Costill et al., 1973; Daniels, 1974; Farrell, et al., 1979; McMiken and Daniels, 1976; Sjödin and Schéle, 1982; Svedenhag and Sjödin, 1984). In elite distance runners with a relatively narrow range in $\dot{V}O_{2\,max}$, the running economy at different speeds has been found to be significantly correlated (r = 0.79 to 0.83) with performance in a 10km race (Conley and Krahenbuhl, 1980). There is also a wide variation in running economy between marathon runners of similar performance capacities (table I). This is illustrated in figure 2 in which the range of oxygen uptake *vs* running velocity is shown for elite runners and good runners (under 3h). The lowest oxygen requirements for specific running velocities were found in the top-class runner Ståhl. His requirement of 59.5 ml/kg/min at 20 km/h was similar to the value of 59.7 ml/kg/min noted for Clayton (Costill et al., 1971). Low oxygen uptakes at submaximal velocities have also been found for other first-rate marathon runners like DeMar (11.3 km/h; Dill, 1965) and Shorter (19.3 km/h; Pollock, 1977). From these reports it appears that a good running economy is an important characteristic of

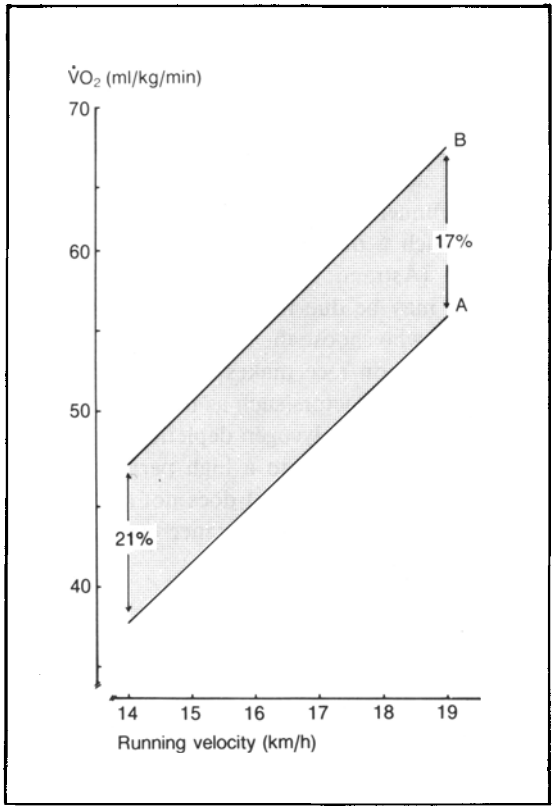

Fig. 2. The shaded area represents the entire range in the oxygen cost of running at different velocities for good and elite runners. A and B represent the regression lines between velocity and oxygen uptake for the runner with the best (Ståhl) and worst running economy, respectively.

elite marathon runners. Further evidence of this is found in the significantly different oxygen costs of running at 15 km/h noted for marathon runners of different performance levels (table I) and also the significant correlation of $r = -0.55$ ($n = 35$, $p < 0.001$) between running economy at 15 km/h and marathon speed (fig. 3). A similar correlation ($r = -0.49$, $p < 0.05$, $n = 13$, 2:17 to 3:49) between marathon performance and oxygen cost of running at 16.1 km/h has been found by Farrell et al. (1979). However, as was the case for $\dot{V}O_{2\,max}$, no significant correlation between running economy and performance is discernible when a subgroup of runners within a narrower performance range in the marathon is studied (Davies and Thompson, 1979) [fig. 3, e.g. for runners between 2:30 and 3:00, $r = 0.25$, n.s., $n = 16$]. Foster et al. (1977) [$n = 23$, range 2:23 to 4:08] also found no significant correlation ($r = -0.36$) between running economy at 13.8 km/h and marathon performance. Nevertheless, it would seem that running economy may be partially related to marathon performance with better economy for elite marathon runners than for slower runners. But the difference is less obvious than that found for $\dot{V}O_{2\,max}$.

On comparing elite runners of different running events but with comparable performance capacities, both Costill (1972) and Pollock (1977) found significantly lower oxygen costs of running in marathon runners than in runners of shorter distances. However, other studies have failed to demonstrate significant differences in this respect (Costill et al., 1973; Svedenhag and Sjödin, 1984). If marathon runners, as a group, indeed have a better running economy than runners of shorter distances, the marathon runners could thus compensate for their

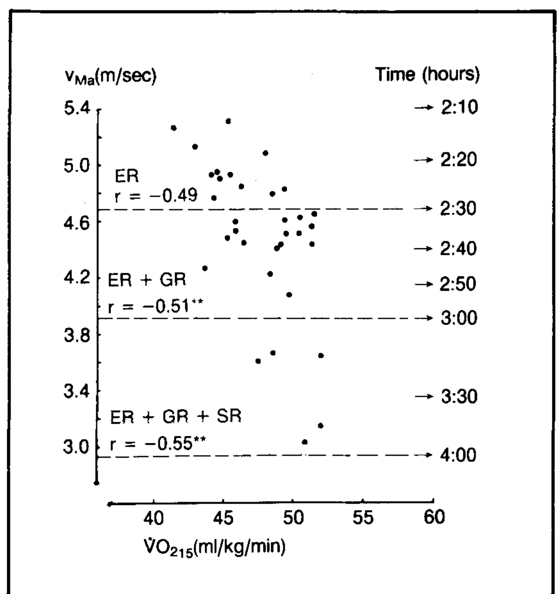

Fig. 3. The relationship between the individual mean marathon velocity (v_{Ma}) and the running economy ($\dot{V}O_{2\,15}$) in ER ($n = 12$), in ER + GR ($n = 28$) and in ER + GR + SR ($n = 35$) ** = $p < 0.01$. For abbreviations see fig. 1.

lower $\dot{V}O_{2\ max}$ values (see above), making their 'total aerobic running capacity' similar to that of the runners of shorter events. In other words, the fractional utilisation of $\dot{V}O_{2\ max}$ when running at a specific submaximal speed (e.g. $\%\dot{V}O_2$ 16.1 $\times$ $\dot{V}O_{2\ max}{}^{-1}$) may be similar (Pollock, 1977). Here again we have no answer to what may be the cause of such a difference in running economy. Thus, if a difference in running economy is accepted, it could either be due to differences in training and/or age or to a selection factor. For instance, long-distance training could lead to a more efficient stride in the marathon runners with small vertical oscillations. As noted above, marathon runners may often be somewhat older than runners of shorter distances (runners often move up the distances gradually). Consequently, more years of training may have resulted in a better running economy. It has been suggested in longitudinal studies that regular training may improve running economy even in adult runners when they start untrained (Conley et al., 1981), but also when they are already well-trained at the time of the first measurements (Svedenhag and Sjödin, 1985). The many factors

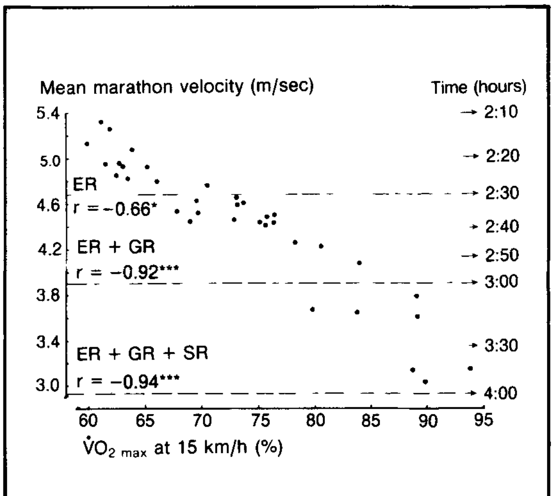

Fig. 4. The relationship between the individual mean marathon velocity (v_{Ma}) and the fractional utilisation of $\dot{V}O_{2\ max}$ at 15 km/h ($\dot{V}O_{2\ 15} \times \dot{V}O_{2\ max} \times 100$) in ER (n = 12), in ER + GR (n = 28) and in ER + GR + SR (n = 35). * = $p < 0.05$; *** = $p < 0.001$. For abbreviations see fig. 1.

Table III. Fractional utilisation of $\dot{V}O_{2\ max}$ at marathon race pace and marathon performance

Authors	n	Mean performance	$\%\dot{V}O_{2\ max}$
Costill et al. (1971)	1	2:08	86
Davies and Thompson (1979)	13	2:13	86
Sjödin and Svedenhag (unpublished data)	12	2:21	80
Costill and Fox (1969)	6	2:27	75
Davies and Thompson (1979)	13	2:30	82
Maughan and Leiper (1983)	5	2:35	74
Sjödin and Svedenhag (unpublished data)	16	2:37	80
Farrell et al. (1979)	13	2:46	75
Wells et al. (1981)	7	3:12	76
Sjödin and Svedenhag (unpublished data)	7	3:24	71
Maughan and Leiper (1983)	5	4:36	60

underlying the oxygen cost of running are still far from being fully understood, however.

2.3 Fractional Utilisation

When the oxygen cost of running at a specific submaximal speed on the treadmill is related to the $\dot{V}O_{2\ max}$ (i.e. $\%\dot{V}O_{2\ max}$), high correlations with performance in endurance events are noted. The fractional utilisation of $\dot{V}O_{2\ max}$ has been found to be significantly correlated with performance at different distances, such as 10 miles (Costill et al., 1973) and 5km (Sjödin and Schéle, 1982). As shown in figure 4, a correlation was also found in the marathon between the fractional utilisation of $\dot{V}O_{2\ max}$ at a submaximal speed (15 km/h, $\%\dot{V}O_{2\ 15} \times \dot{V}O_{2\ max}{}^{-1}$) and performance capacity ($r = -0.94$, $n = 35$, $p < 0.001$). The clearly different values of the different running categories illustrate this further (table I). These good correlations between $\%\dot{V}O_{2\ max}$ at a specific submaximal speed and performance are attributable to the fact that this

$\%\dot{V}O_{2\ max}$ value expresses both the effects of $\dot{V}O_{2\ max}$ and of running economy, which may both be separately related to performance.

Another aspect of fractional utilisation is the question of how close to $\dot{V}O_{2\ max}$ the runner can perform during competition. Obviously, this value is dependent on the distance (Costill and Fox, 1969) as well as on environmental factors (see below). This $\%\dot{V}O_{2\ max}$ at the marathon race pace ($\%\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1}$, determined on the treadmill) has been found to range between 60% in slow runners to 86% in elite runners (table III). Our data shows that the $\%\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1}$ was the same, or 80%, in both elite runners (mean 2:21) and good runners (2:37) [table 1]. However, in the group exceeding 3h, the value was significantly lower (71%). When the $\%\dot{V}O_{2\ max}$ at race pace has been correlated with marathon performance, both significant ($r = 0.74$, $n = 18$, 2:19 to 4:53, Maughan and Leiper, 1983) and non-significant ($r = 0.07$, $n = 13$, 2:17 to 3:49, Farrell et al., 1979) correlations have been found. In our data, a significant correlation was noted only when the complete data was considered ($r = 0.70$, $p < 0.01$, $n = 35$) [fig. 5]. Thus, in the runners with times faster than 3h ($n = 28$) no correlation was found ($r = 0.09$). These findings suggest that the $\%\dot{V}O_{2\ max}$ that can be maintained during a marathon race differs in runners showing large differences in performance capacity. In experienced runners prepared for the event with adequate endurance training (such as the sub-3-hour runners in our data; see below) the $\%\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1}$ may be very similar. The lower $\%\dot{V}O_{2\ max}$ in the slowest runners may not only be due to a lack of adequate endurance training, however, but is probably also related to the longer period of exertion for these runners.

Costill and Winrow (1970) noted that age may be one of the factors determining the $\%\dot{V}O_{2\ max}$ at the marathon race pace. Accordingly, their younger runners employed about 65 to 75% of their $\dot{V}O_{2\ max}$ while their older runners could utilise 80 to 85% of $\dot{V}O_{2\ max}$ when running the marathon. To study this further, the sub-3-hour runners in our data were divided into a younger and an older group, with a mean difference of 10 years (younger group, $n = 13$, 22.7 years, range 18-27; older group, $n = 15$, 32.7 years, range 28-44). Although the difference in $\%\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1}$ determined in this way was not as large as that noted by Costill and Winrow (1970), the older runners were able to use a significantly ($p < 0.05$) higher percentage of their $\dot{V}O_{2\ max}$ during the marathon (81.4 ± 0.5%) than the younger runners (78.8 ± 1.0%). However, age appears to be clearly less important than adequate endurance training in raising the $\%\dot{V}O_{2\ max}$ because a lower value was noted for the runners with times over 3h in spite of the fact that they were older than both the elite and good runners. An alternative explanation, as suggested also by Costill and Winrow (1970), is that it is not the age *per se* that is important for the $\%\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1}$, but rather the number of years of regular endurance training.

In all of the above-cited studies the $\%\dot{V}O_{2\ max}$ at race pace was calculated from oxygen uptake determinations on the treadmill. Only one study (to our knowledge) has actually determined the oxygen

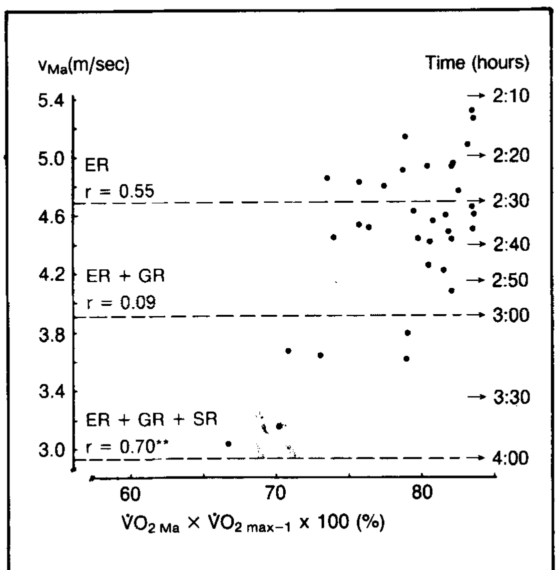

Fig. 5. The relationship between the individual mean marathon velocity and the fractional utilisation of $\dot{V}O_{2\ max}$ at the marathon race pace ($\dot{V}O_{2\ Ma} \times \dot{V}O_{2\ max}^{-1} \times 100$) in ER ($n = 12$), in ER + GR ($n = 28$), in ER + GR + SR ($n = 35$). ** = $p < 0.01$. For abbreviations see fig. 1.

uptake during a marathon race. In this study, Maron et al. (1976) found that the $\%\dot{V}O_{2\,max}$ in 2 marathon runners (2:36 and 2:39) varied between 68% and 100%, depending on whether they were running downhill or uphill. When running on the level they utilised 77% of their $\dot{V}O_{2\,max}$. If the effect of air resistance on the runners is allowed for to facilitate a comparison with the $\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$ values noted above, the corresponding value would be between 73% and 75% of the $\dot{V}O_{2\,max}$ for running on the level (Davies, 1980; Pugh, 1970).

2.4 'Anaerobic Threshold'

In recent years it has been suggested that metabolic parameters measured at submaximal exercise may be better indicators for endurance exercise capacity than maximal oxygen uptake measurements. Different terms such as 'anaerobic threshold', 'aerobic threshold' and 'onset of blood or plasma lactate accumulation' (OBLA and OPLA) have been introduced by different authors for the corresponding submaximal measurements of threshold velocity. These have been determined from ventilatory parameters (Wasserman et al., 1973), lactate values for blood or plasma (2.2 mmol/L: Londeree and Ames, 1975; 2 and 4 mmol/L respectively: Kindermann et al., 1979; increase in delta lactate: Farrell et al., 1979) as well as from heart rate measurements (Conconi et al., 1982). These terms are commonly used in spite of the fact that all the mechanisms underlying the different threshold concepts have not been clarified. Nevertheless, there seems to be a rather close relationship between these 'thresholds' and performance. Thus, in long-distance running, high correlations of the order of r = 0.88 to 0.99 have been reported for 3.2km to the half marathon (Conconi et al., 1982; Farrell et al., 1979; Powers et al., 1983; Sjödin and Schéle, 1982; Williams and Nute, 1983). In the marathon, correlations of the order of r = 0.94 to 0.98 have been found between threshold speed and marathon velocity (table IV). In the present material, a correlation of r = 0.96 was noted between the 4 mmol/L speed ($V_{Hla\,4}$) and the marathon velocity (fig. 6). In this case, as opposed to the variables presented above, there were also significant

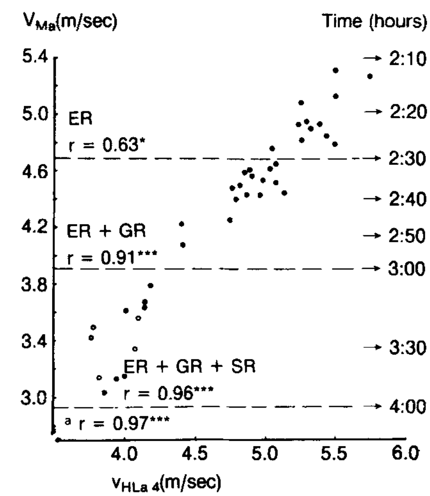

Fig. 6. The relationship between the individual mean marathon velocity and the running velocity corresponding to the anaerobic threshold ($v_{Hla\,4}$) in ER (n = 12), in ER + GR (n = 28) and in ER + GR + SR [n = 35 + 6 extra runners ([a]) in the SR group]. * = $p < 0.05$; *** = $p < 0.001$.

Table IV. Correlations between 'anaerobic threshold' velocity and marathon performance

Authors	n	Performance range	Correlation
Farrell et al. (1979)	13	2:17-3:49	r = 0.98
Sjödin and Jacobs (1981)	18	2:22-4:12	r = 0.96
Conconi et al. (1982)	55	About 2:10-3:00	r = 0.95
Rhodes and McKenzie (1984)	18	2:13-3:21	r = 0.94
Sjödin and Svedenhag (unpublished data)	35	2:12-3:52	r = 0.96

All correlations were significant.

correlations between the 4 mmol/L velocity and performance in all the subgroups of runners (e.g. for runners under 2:30, r = 0.63; 2:30 to 3:00, r = 0.76; over 3:00, r = 0.92). In addition, in the ANOVA for the 3 different groups of runners in table I, the highest F-ratio was found for the threshold variable. Thus, it appears from this that the 'threshold' is the best single predictor of performance in long-distance running, including the marathon. This is logical since the threshold is dependent on both the $\dot{V}O_{2\,max}$ and running economy as well as on the fractional utilisation of $\dot{V}O_{2\,max}$.

The $\%\dot{V}O_{2\,max}$ at the marathon race pace ($\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$), as discussed above, can be divided into 2 subordinate factors:

1) The percentage of the 'anaerobic threshold' velocity utilised while running the marathon, and

2) The percentage of the $\dot{V}O_{2\,max}$ utilised while running at this 'threshold'. Considering the former, we found that the elite and good runners ran at similar levels, or 92 to 93% of the 4 mmol/L velocity ($V_{Hla\,4}$). (Examples of how runners in these 2 groups performed in their races relative to their thresholds are shown in figure 7 in which the mean curve of 7 runners (2:24 to 2:45) and the individual curves of 2 extreme runners are shown.) The runners over 3h ran at a significantly lower level, or 85% of the 4 mmol/L velocity, however (table I). This may suggest that runners of clearly different abilities in the marathon run at different percentages of the anaerobic threshold velocity. In concordance with this, Sjödin and Jacobs (1981) noted a significant correlation (r = 0.86) between % threshold velocity at marathon race pace and performance (n = 18, 2:22 to 4:12). In our unpublished data, a significant correlation was found for the whole group (r = 0.76, n = 35, $p < 0.001$) but not for the sub-3-hour runners (r = 0.26, n = 28, n.s.). The reasons for the lower percentage of the 4 mmol/L velocity for slow marathon runners could be due to their longer running times, or to a lower capacity to save glycogen and use fat (see below), or just to inadequate experience in running, since in the study of Farrell et al. (1979) no relationship between percentage of anaerobic threshold and marathon performance was seen in experienced runners (n = 13, 2:17 to 3:49). In the study by Farrell et al. (1979) the marathon race pace relative to the anaerobic threshold was approximately 103%, which is much higher than the values we noted above. The difference is probably due to the fact that their anaerobic threshold (OPLA) was determined from the first increase in plasma lactate while we employed the 4 mmol/L blood lactate criterion.

The second subordinate factor to the $\%\dot{V}O_{2\,max}$ at the marathon race pace is the $\%\dot{V}O_{2\,max}$ while running at the 'anaerobic threshold'. This value was similar (88%) in the 2 sub-3-hour groups but was slightly, although significantly, lower (85%) in the slow runners (table I). With physical endurance training, the lactate curve is known to be shifted to the right relative to $\dot{V}O_{2\,max}$, resulting in lower lactate levels at the same $\%\dot{V}O_{2\,max}$ (at least in the 60% to 90% range) [Costill, 1970; Hurley et al., 1984; MacDougall, 1977). In the latter study by Hurley et al. the lactate curve was significantly shifted to the right after 12 weeks of training in previously untrained subjects. We have observed previously that the $\%\dot{V}O_{2\,max}$ at $V_{Hla\,4}$ was not significantly different between marathon runners

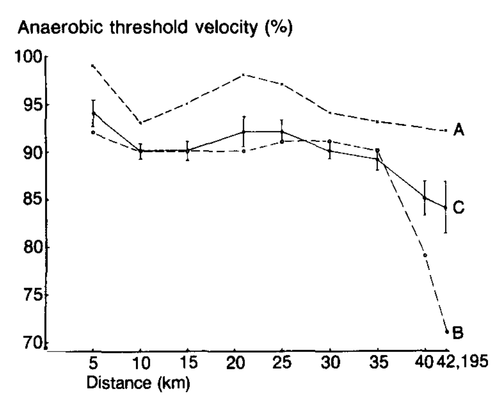

Fig. 7. Running velocity during marathon expressed in percentage of the 'anaerobic threshold' velocity ($V_{Ma} \times V_{HLa}\,4^{-1} \times 100$) in a group of elite and good runners, C (means and standard error, n = 7). In addition, the individual curves for the two runners, A and B, who deviated most from the mean curve (C), are given.

and runners in the 400m and 800m events (Svedenhag and Sjödin, 1984). Although the $\%\dot{V}O_{2\,max}$ at the 4 mmol/L velocity differed significantly between sub-3-hour runners and slower runners in our unpublished data, the difference was not marked. In all, this may indicate that the rightward shift of the lactate curve relative to $\dot{V}O_{2\,max}$ may occur largely as an early response to training. The excessive amounts of training often done by marathon runners thus appear to have a relatively small effect on further shifting of the lactate curve to the right.

Training at the speed corresponding to the anaerobic threshold is widely used as a means of improving performance in marathon running (Lenzi, 1983). In view of the discussion presented above concerning the early rightward shift of the lactate curve, such a training effort would appear to be rather fruitless for a previously trained individual. However, an improvement in the anaerobic threshold velocity after threshold training in previously well-trained runners may be mostly due to effects on the running economy and the $\dot{V}O_{2\,max}$ (Sjödin et al., 1982). Evidently, there may be an intricate interplay between training and the different physiological variables determining running performance.

As was the case for $\%\dot{V}O_{2\,max}$ at the marathon race pace, $\%\dot{V}O_{2\,max}$ at the 4 mmol/L velocity was significantly higher in the older (sub-3-hour) runners than in the younger sub-3-hour runners (89.3 $\pm$ 0.6% *vs* 86.8 $\pm$ 0.6%). The highest value was found in the 35-year-old runner Ståhl (92.9%). This may explain, then, why the older runners were able to utilise a higher $\%\dot{V}O_{2\,max}$ at the marathon race pace than the younger runners.

2.5 Fuels

2.5.1 Energy Expenditure

The total energy expenditure for completing the 42,195 metres of a marathon race has been calculated to amount to 9000 to 12,000 kJ (Costill, 1972; Newsholme, 1983; Wells et al., 1981). Some deviation between runners may result from individual variations in the running economy. For elite marathoners who complete the race in a faster time than 2:30, this means that the rate of energy expenditure may exceed 75 kJ/min during the race (Costill and Fox, 1969; Sjödin and Svedenhag, unpublished data). One problem for the body is to meet this huge energy demand with an adequate amount of efficient fuel.

Available fuels are carbohydrates and lipids and, to a smaller extent, proteins (Dohm et al., 1982). Endogenous fuels, such as muscle glycogen and muscle lipid stores, seem to be utilised more efficiently by the contracting muscle than exogenous energy sources such as circulating free fatty acids (FFA) and blood glucose (Felig and Wahren, 1975; Lithell et al., 1979). However, as the performance progresses and the endogenous sources become more depleted, the circulatory substrates glucose and FFA gradually become more important (Felig and Wahren, 1975; Lithell et al., 1979).

To make a fair estimate of the utilisation of different fuels by marathon runners, one must consider the high-power output (up to 85% of $\dot{V}O_{2\,max}$) that can be maintained by the elite runners throughout the race (see above). At correspondingly high exercise intensities we know that there is a greater reliance on carbohydrate than on lipids. Thus, in moderately trained subjects the rate of muscle glycogen utilisation at least above 50% of $\dot{V}O_{2\,max}$ has been found to be directly proportional to the exercise intensity (Hermansen et al., 1967).

2.5.2 Glycogen Stores

The mean glycogen concentration in the leg muscles has been found to be about 12 g/kg wet muscle, but there are large intraindividual variations (Hultman, 1967). Somewhat higher mean values, 17 to 18 g/kg wet muscle, have been demonstrated in endurance-trained individuals (Hermansen et al., 1967; Karlsson and Saltin, 1971). Assuming that the active muscle mass amounts to about 20kg, the total available muscle glycogen store should amount to 340 to 360g. By depleting the glycogen stores by exercise combined with a very low carbohydrate intake for 1 or 2 days and then replenishing the stores by consuming a carbohydrate-rich diet for another 2 or 3 days, the muscle glycogen stores can be more than doubled (Bergström et al., 1967). Calculations have been published which indicate that the degradation of muscle glycogen proceeds at a rate of 0.4 to 0.5 g/kg/min wet muscle tissue at exercise intensities similar to those in marathon running, i.e. 85% to 90% $\dot{V}O_{2\,max}$ (Karlsson and Saltin, 1971). Muscle glycogen consumption may be even higher after the carbohydrate loading procedure (Costill and Miller, 1980). If the muscle glycogen stores were used exclusively, these stores ought to be depleted within 40 minutes with normal stores or within 70 minutes with enlarged stores. An additional 100g glycogen may be stored in the liver and another 20g glucose is present in extracellular fluids (Newsholme, 1981). Theoretically, if all these carbohydrates could be transferred and utilised by the exercising muscles, these exogenous stores could delay carbohydrate depletion by another 15 minutes.

2.5.3 Lactate Oxidation

In addition, some extra glucose may be produced from lactate and some amino acids by gluconeogenesis in the liver (Felig and Wahren, 1975). Even though relatively low lactate values have been found after a marathon race (2.1 mmol/L, n = 6, Costill and Fox, 1969; 1.9 mmol/L, n = 6, Maron et al., 1975), higher values may occur during uphill parts of the course as the runner may then be performing at 100% of his or her $\dot{V}O_{2\,max}$ (Maron et al., 1976). Lactate can probably also be used directly in the active musculature. Thus, the alternative pathway in which lactate is oxidised is catalysed by the H-LDH isozyme (Skilleter and Kun 1972). As mentioned, relatively high activities of this isozyme have been found in the type I fibres of endurance-trained individuals (Sjödin, 1976), indicating a high capacity for lactate oxidation.

2.5.4 Glycogen Depletion and Carbohydrate Supplementation

By making similar assumptions about the delaying effect on glycogen depletion resulting from the availability of other carbohydrate sources, Locksley (1980) has calculated the total muscle glycogen depletion will occur at between 32 and 40km of the marathon race. A corresponding situation is illustrated in figure 7. Runner B obviously 'hit the wall' 7km from the finish and was thoroughly exhausted during the last 5km. By administering additional oral carbohydrate solutions during the race such a depleted condition might have at least been delayed (Coyle and Coggan, 1984). The emptying rate of the stomach is limited, however, particularly during intense exercise. Accordingly, the emptying rate is inversely related to the intensity of the exercise and the glucose concentration of the solution (Costill and Saltin, 1974). Too high a concentration of monosaccharides in the solution will produce an excessively high osmotic pressure in the gastrointestinal tract, and this will result in delayed gastric emptying and slower uptake from the intestine. This could be avoided, however, with a low concentration of the monosaccharides (glucose and fructose) or by using polysaccharides in the drinks (Daum et al., 1978).

2.5.5 Fat Metabolism

From the above discussion it is clear that the stores of carbohydrate in the body are small and will not cover the energy demands of running a marathon. As illustrated above, there is considerable evidence that carbohydrate depletion may be the main factor responsible for exhaustion during prolonged exercise. If the stores are depleted, the exercise intensity has to be reduced. However, some authors have shown that even with minimal muscle glycogen stores the exercise can still be performed at a power output of 60 to 70%, or less, of $\dot{V}O_{2\,max}$, provided that the supply of free fatty acids is adequate (Pernow and Saltin, 1971). If the supply of FFA is adequate from the beginning of exercise, this may have a sparing effect on the muscle glycogen stores. It has been demonstrated that if the FFA concentration is artificially elevated, e.g. by an intake of caffeine before exercise, a specific level of exercise can be maintained for a longer period of time (Costill et al., 1977; Hickson et al., 1977).

One of the major adaptations to endurance training is probably also related to an increased ability to oxidise fat in the active muscle tissue.

This is indicated by elevated activity levels of fat-oxidising enzymes in the skeletal muscle of endurance-trained individuals as discussed earlier (Chi et al., 1983; Essén-Gustavsson and Henriksson, 1984). Respiratory data on marathon runners have also demonstrated that more than 75% of the consumed energy could be derived from fat while running at 70% of $\dot{V}O_{2\,max}$ (Costill et al., 1979). This indicates a considerably higher turnover rate of fat metabolism than can be expected in a normal population at correspondingly high relative exercise intensities. All these metabolic adaptations may make it possible to delay the depletion of carbohydrate stores and thereby contribute to the ability of elite marathon runners to maintain a very high and constant running speed during the whole race because the most efficient fuels, the carbohydrates, will be available to the very end of the race.

2.6 Environmental and Other Factors

In addition to the physiological factors discussed above, marathon performance is dependent on several other factors. These factors are, in part, determinants of the velocity and therefore also of the $\%\dot{V}O_{2\,max}$ that can be maintained during the race (see above). Thus, the physical characteristics of the marathon course (e.g. hilly or flat), the altitude of the course and whether there is a difference in altitude between the start and finish lines are obviously factors of importance to marathon performance. Other environmental factors of importance include the ambient temperature and humidity (Costill, 1972) and the wind conditions during the race. Furthermore, body fluid losses and their replacement, together with the runner's footwear and clothing (Costill and Fox, 1969), may also affect performance as well as psychological characteristics (including motivation) of the runner. The latter may be true even though marathon runners as a group have been found to score within normal limits of most psychological variables (Morgan and Costill, 1972).

The matter of body fluid losses and their replacement deserves some further notice. The bodyweight losses during a marathon may be quite large, or up to 6kg (Costill, 1972; Pugh et al., 1967). Wyndham and Strydom (1969) showed that the rectal temperature of the runners will rise when the water deficit (primarily due to perspiration) exceeds 3% of the bodyweight, even when running under cool conditions (less than 17°C). Maron et al. (1977) suggested decreased sweating as the cause of the increase in rectal temperature. The rectal temperature of marathon runners may, in fact, exceed 41°C during the latter part of a race (Costill, 1972; Maron et al., 1977; Pugh et al., 1967). Even though such high rectal temperature may not always impair performance (Maron et al., 1977), the risk is considerable (Costill, 1972). This points to the importance of fluid replacement during a marathon race in order to prevent hyperthermia, especially when the race is run under warm conditions. Even though only partial replacement of the fluid loss is possible, fluid intake has been shown to suppress the increase in rectal temperature with running time (Costill, 1972).

3. Training and Marathon Performance

To be able to run a marathon race at maximal individual capacity, the runner must be sufficiently prepared for the race. The most important part of this preparation appears to be the pre-marathon training in the months immediately preceding the event. Of the different training indices employed by long-distance runners, the mean distance per week during the 2 months preceding the race probably shows the highest correlation with marathon performance (Hagan et al., 1981). Thus, significant correlations of $r = 0.94$ ($n = 18$, 2:22 to 4:12, Sjödin and Jacobs, 1981) and $r = 0.67$ ($n = 50$, 2:19 to 4:58, Hagan et al., 1981) have been found between average weekly kilometrage during the last 2 months and marathon speed. In our data, a value of $r = 0.85$ ($p < 0.001$) was found for the whole series ($n = 35$) but significant correlations were also found within the different subgroups (fig. 8). These high correlations between marathon performance and training kilometrage are to be expected in runner populations with ranges of performance capacities as wide as the ones presented above. Still, there

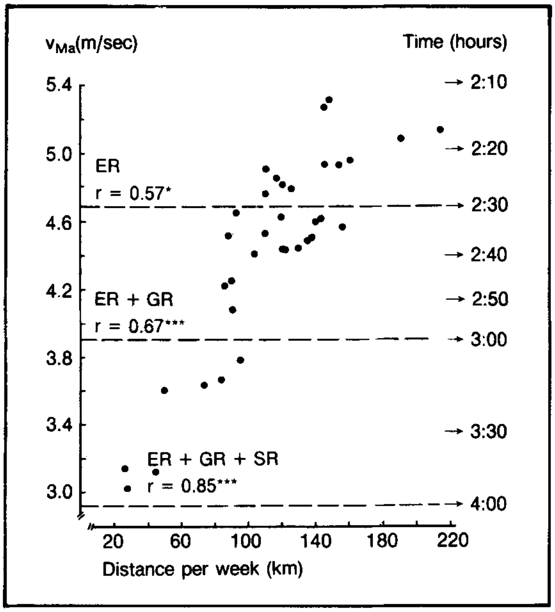

Fig. 8. The relationship between the individual mean marathon velocity (v_{Ma}) and the training kilometrage per week in ER (n = 12), ER + GR (n = 28) and ER + GR + SR (n = 35). * = $p < 0.05$; *** = $p < 0.001$.

would probably be an even better correlation between training and performance if it were possible to study the effect of training in a group of runners with similar natural talent.

Some interesting observations can be made when studying the different categories of runners in table I. As can be seen, the 3 groups differed significantly in their average weekly kilometrage during the 10 weeks preceding the marathon. Thus, the values for the elite, good and slow runners were 145, 115 and 57 km/week, respectively. The relatively small amount of long-distance training by the slow runners could partly explain their low $\dot{V}O_{2\,max}$, but it may account to a much larger extent for the lower $\%\dot{V}O_{2\,max}$ utilised by these runners during the marathon ($\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$) [see above]. The improvement from 3:20 to 3:10h noted by Foster et al. (1977) when the training of 12 runners was increased from 76 to 91 km/week may accordingly be due to an improvement in the $\%\dot{V}O_{2\,max}$ maintained during the race. As has also been discussed by these authors, this extra long-distance training is likely to have caused an increased oxidative capacity of skeletal muscle which should have resulted in an increased ability to oxidise fat and lactate (see above).

However, the fractional utilisation of $\dot{V}O_{2\,max}$ by elite and good runners during the running of the marathon was equal despite the fact that the elite runners trained 30km more each week ($p < 0.05$). Thus, there may be an upper limit of training distance above which there is no further increase in the $\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$. If this is true, what is the point of having an average weekly kilometrage in excess of 115 to 120km? Most elite marathoners nowadays seem to have a training distance which exceeds this value. First of all, it is doubtful whether such an extra training distance above, let us say, 120 km/week affects the $\dot{V}O_{2\,max}$. The $\dot{V}O_{2\,max}$ is relatively stable in well-trained athletes (Åstrand and Rodahl, 1977; Bergh, 1982) and the small fluctuations per kg bodyweight that might occur in middle- and long-distance runners during a year may be related more to the intense training during the competitive period than to training distance (Svedenhag and Sjödin, 1985). In this context, it may also be relevant to recall the above discussion of the cause of the suggested lower $\dot{V}O_{2\,max}$ values in marathoners compared with long-distance runners in shorter events.

Consequently, the difference between $\dot{V}O_{2\,max}$ in good and elite runners would be the result of differences in natural talent. Secondly, hypothetically, the extra training distance could be a way to keep bodyweight down and thus performance up, but no difference in bodyweight was seen between elite and good runners. If this extra distance each week has a beneficial effect at all (which can be questioned), it might be due to a positive effect on running economy. As discussed above, it has been suggested that regular training may improve running economy even in adult and previously well-trained runners (Svedenhag and Sjödin, 1985). Thus, the higher training distance could hypothetically hasten the development of a better running economy. If exaggerated, however, the running economy may be improved only at the slower training speed and not, as intended, at the racing speed. It must be remembered, though, that to an elite marathon runner, a high training distance may be important for other reasons than solely to increase performance capacity in a single race. It can be speculated that a high training kilometrage may shorten the recovery period after races, reduce the risk of getting injured during competition and intensive training and better preserve good form. These factors may, of course, be of great importance to ambitious runners who want to compete often.

4. Conclusion

It is safe to say that marathon runners must rely to a large extent on a high aerobic capacity, expressed as the maximal oxygen uptake ($\dot{V}O_{2\,max}$). There are, however, great variations in $\dot{V}O_{2\,max}$ among elite runners with similar performance capacities. Thus, an estimation made by stepwise multiple regression analysis showed that only 61% of the variation in performance capacity could be explained solely by $\dot{V}O_{2\,max}$ in our unpublished material (n = 35).

It is also suggested that a low $\dot{V}O_{2\,max}$ can be compensated for by good running economy. This is measured as oxygen uptake at submaximal running velocities determined in our laboratory at 15 km/h($\dot{V}O_{2\,15}$). By adding this variable to $\dot{V}O_{2\,max}$ the predictive power for marathon performance increased significantly to 84% in the same group of runners. Another factor of importance is the fractional utilisation of $\dot{V}O_{2\,max}$ at race pace ($\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$). Together, $\dot{V}O_{2\,max}$, $\dot{V}O_{2\,15}$ and $\%\dot{V}O_{2\,Ma} \times \dot{V}O_{2\,max}^{-1}$ could explain 98% of the variation in the marathon performance of our group. To a corresponding extent (93%), these variables also explain the variation in the 'anaerobic threshold', a factor which is related to the metabolic response to increasing exercise intensities. From that it is obvious that the anaerobic threshold is the single variable which has the highest predictive power for marathon performance. Thus, 92% of the variation in performance was explained solely by this factor among our runners.

A major limiting factor to marathon performance, which is related to the fractional utilisation of $\dot{V}O_{2\,max}$ at race pace, is probably the choice of fuels for the contracting muscles. Published data indicate a considerably higher turnover rate in fat metabolism at relatively high exercise intensities (70% of $\dot{V}O_{2\,max}$) in marathon runners compared with untrained individuals. Since the stores of the most efficient fuel, carbohydrate, are limited, the selection of fat for oxidation by the exercising muscles is of considerable importance for delaying the onset of fatigue. The large amount of endurance training done by the marathoners is probably responsible for this adaptation in order to utilise fat even at high exercise intensities.

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